Name: Jakob Tomas West u0675702

Lab07

Professor Nabil cs6015

Q1: There were no results, surely no functions were ran.

```
Flat profile:

Each sample counts as 0.01 seconds.
 no time accumulated

  %   cumulative   self              self     total
 time   seconds   seconds    calls  Ts/call  Ts/call  name
  0.00      0.00     0.00     8136     0.00     0.00  __gnu_
g<char, std::char_traits<char>, std::allocator<char> > const
  0.00      0.00     0.00     8136     0.00     0.00  bool s
r_traits<char>, std::allocator<char> > const&, std::__cxx11:
  0.00      0.00     0.00     5686     0.00     0.00  bool s
r_traits<char>, std::allocator<char> > const&, std::__cxx11:
  0.00      0.00     0.00     1026     0.00     0.00  std::c
  0.00      0.00     0.00      149     0.00     0.00  search
string<char, std::char_traits<char>, std::allocator<char> >)
  0.00      0.00     0.00      141     0.00     0.00  void s
ts<char>, std::allocator<char> >&, std::__cxx11::basic_strin
  0.00      0.00     0.00        2     0.00     0.00  find_p
11::basic_string<char, std::char_traits<char>, std::allocato
  0.00      0.00     0.00        2     0.00     0.00  sort1(
  0.00      0.00     0.00        2     0.00     0.00  readFi
g<char, std::char_traits<char>, std::allocator<char> >*)
  0.00      0.00     0.00        1     0.00     0.00  _GLOBA
  0.00      0.00     0.00        1     0.00     0.00  _GLOBA
  0.00      0.00     0.00        1     0.00     0.00  __stat
  0.00      0.00     0.00        1     0.00     0.00  __stat
```

Q2:

```
index % time    self  children    called     name

[1]     99.6    0.00    3.56                 main
                0.00    2.55       2/2
ar>, std::allocator<char> >*, int, int, std::__cx
                0.66    0.35       2/2
                0.00    0.00       2/2
ator<char> >*) [18]
-----------------------------------------------
                0.63    1.92   38948/38948
ar>, std::allocator<char> >*, int, int, std::__cx
[2]     71.3    0.63    1.92   38948         sear
cator<char> >) [2]
                0.42    1.50 647482750/647482750
 const&, std::__cxx11::basic_string<char, std::ch
-----------------------------------------------
                0.00    2.55       2/2
[3]     71.3    0.00    2.55       2         find
 std::allocator<char> >*, int, int, std::__cxx11:
                0.63    1.92   38948/38948
allocator<char> >) [2]
-----------------------------------------------
                0.42    1.50 647482750/647482750
td::allocator<char> >) [2]
[4]     53.6    0.42    1.50 647482750         bo
&, std::__cxx11::basic_string<char, std::char_tra
                1.19    0.31 647482750/647482750
llocator<char> > const&, std::__cxx11::basic_stri
-----------------------------------------------
                1.19    0.31 647482750/647482750
 const&, std::__cxx11::basic_string<char, std::ch
[5]     41.9    1.19    0.31 647482750         __
or<char> > const&, std::__cxx11::basic_string<cha
                0.31    0.00 108224639/108224639
-----------------------------------------------
                0.66    0.35       2/2
[6]     28.3    0.66    0.35       2         sort
                0.35    0.00 379465206/379465206
 const&, std::__cxx11::basic_string<char, std::c
                0.00    0.00   38931/38931
cxx11::basic_string<char, std::char_traits<char>
-----------------------------------------------
                0.35    0.00 379465206/379465206
[7]      9.8    0.35    0.00 379465206         b
&, std::__cxx11::basic_string<char, std::char_tr
-----------------------------------------------
                0.31    0.00 108224639/108224639
llocator<char> > const&, std::__cxx11::basic_str
[8]      8.5    0.31    0.00 108224639         s
-----------------------------------------------
                0.02    0.00       1/1
[9]      0.4    0.02    0.00       1         _GL
                0.00    0.00       1/1
-----------------------------------------------

[10]     0.4    0.00    0.02                 __l
                0.02    0.00       1/1
                0.00    0.00       1/1
-----------------------------------------------
                0.00    0.00   38931/38931
[17]     0.0    0.00    0.00   38931         voi
1::basic_string<char, std::char_traits<char>, st
-----------------------------------------------
                0.00    0.00       2/2
[18]     0.0    0.00    0.00       2         rea
<char> >*) [18]
-----------------------------------------------
                0.00    0.00       1/1
[19]     0.0    0.00    0.00       1         _GL
                0.00    0.00       1/1
-----------------------------------------------
                0.00    0.00       1/1
[20]     0.0    0.00    0.00       1         __s
-----------------------------------------------
                0.00    0.00       1/1
[21]     0.0    0.00    0.00       1         __s
-----------------------------------------------
```

```
Flat profile:

Each sample counts as 0.01 seconds.
  %   cumulative   self              self     total
 time   seconds   seconds    calls   s/call   s/call  name
 33.41      1.19     1.19 647482750     0.00     0.00  __gnu_cxx::__enab
:allocator<char> > const&, std::__cxx11::basic_string<char, std::char_tr
 18.53      1.85     0.66        2     0.33     0.51  sort1(std::__cxx11
 17.69      2.49     0.63    38948     0.00     0.00  search1(std::__cxx
std::allocator<char> >)
 11.79      2.91     0.42 647482750     0.00     0.00  bool std::operato
 > const&, std::__cxx11::basic_string<char, std::char_traits<char>, std:
  9.83      3.26     0.35 379465206     0.00     0.00  bool std::operato
 > const&, std::__cxx11::basic_string<char, std::char_traits<char>, std:
  8.56      3.56     0.31 108224639     0.00     0.00  std::char_traits<
  0.42      3.58     0.02        1     0.02     0.02  _GLOBAL__sub_I__Z7
  0.00      3.58     0.00    38931     0.00     0.00  void std::swap<cha
d::__cxx11::basic_string<char, std::char_traits<char>, std::allocator<ch
  0.00      3.58     0.00        2     0.00     1.28  find_print_add_rec
ts<char>, std::allocator<char> >*, int, int, std::__cxx11::basic_string<
  0.00      3.58     0.00        2     0.00     0.00  readFile(std::__cx
allocator<char> >*)
  0.00      3.58     0.00        1     0.00     0.00  _GLOBAL__sub_I__Z8
  0.00      3.58     0.00        1     0.00     0.00  __static_initializ
  0.00      3.58     0.00        1     0.00     0.00  __static_initializ
```

Q3: From the flat profile, identify which function consumes the highest time percentage of the program execution time? Report the percentage.

Ignoring gnu_cxx, The user defined function Sort1 had the highest time percentage of program execution. The percent time was 18.53%.

Q4: Deduce which function is the bottle-neck in the program?

Sort1 seems to be the bottle neck in the program.

Q5: From the flat profile, record the seconds per call (self and total) / (Weight and self-weight in Xcode - time profiler) for `search1` function. Explain the values of self and total seconds per call.

The record for self and total per second was **0.00 seconds**. The search1 took so little time it couldn't be rounded up to the hundredths place. It's probably a small function and doesn't call other functions. The total call includes the child time.

Q6: Find,Print, and add functions. The self time was **0.00** seconds but when it's calling other functions it adds to the total time resulting in an increase in time of **1.02** seconds.

Q7: Record the child function that is contributing most to the time of main function. Indicate the total amount it propagates to main.

The total time was 2.04 seconds contributing from the find_print_add_records child function.

Q8: Profile the program. Write in the results for different optimizations.

Optimization Level	Total execution time
O0 (default)	3.30 seconds
O1	2.89 seconds
O2	1.97 seconds
O3	1.64 seconds

Q9: How did the time percentages of each of the program functions change for the three different optimization levels? Interpret

Optimization Level	Search1	Sort1	
O0 (default)	71.3%	28.3%	
O1	70.8%	28.8%	
O2	56.3%	43.75%	
O3	50.6	49.4	

Q10: What is the best optimization for minimizing the execution time?

Optimization3 was the best for minimizing the execution time.

Q11:

Sort function	Search function	Total execution time
Sort1	Search1	3.36 seconds
Sort2	Search1	4.72 seconds
Sort3	Search1	2.42 seconds
Sort1	Search2	0.84 seconds
Sort2	Search2	2.76 seconds
Sort3	Search2	No time propagated

Q12: Sort1 Search2 was the best performance in terms of execution time for me but, others who got a time propagated Sort3 Search2 was the best performance.

Q13: Calculate the program enhancement percentage by the new function optimization?

(3.36 - 0.84)/3.36 = .75 *100 = 755 program enhancement percentage.